

E.3 / PAPER 1 / QUESTION 26

2 May 2023 / TZ2 (file label) / HL Paper 1 / original Q22 / 2223-6513

26. A student measures the count rate of a radioactive sample with time in a laboratory. The background count in the laboratory is 30 counts per second.

Count rate / counts per second	Time / s
150	0
90	20

What is the time at which the student measures a count rate of 45 counts per second?

- A. 30 s
- B. 40 s
- C. 60 s
- D. 80 s